

Lab Overview

The field of Electronics Engineering is characterized by rapid and continuous evolution in both philosophy and technology. Among the essential tools for workforce development, the lab manual plays a vital role. While textbooks provide a theoretical foundation on analytical themes in electronics engineering, lab manuals that present practical systems are necessary for developing both intellectual and motor skills in young diploma engineers. The low voltage requirements of electronic devices, coupled with their ability to control high-power electrical equipment, have made the use of electronic devices increasingly prominent in the electrical industry. With the evolution of integrated circuits, the space requirements for these devices have also decreased. Electronic control methods surpass other control techniques in terms of cost, reliability, maintenance, and power management. Therefore, studying basic electronic components equips individuals with the necessary tools for the electrical control industry.

Objectives:

1. Understand the basic principles of electronic devices.
2. Identify and specify various electronic components and devices.
3. Understand the construction and working principles of different electronic devices.
4. Understand the use of different electronic devices.

Contents:

Lab 1: Introduction to Electronics Laboratory

Lab 2: DC Power Supply

Lab 3: Common Emitter Amplifier

Lab 4: MOSFET Amplifier

Lab 5: Operational Amplifiers

Introduction to lab equipment

The basic equipment found in an electronic laboratory includes a power supply, function generator, oscilloscope, multimeter, and breadboard.

- Power Supply: This device generates variable DC voltages for use in circuits.
- Function Generator: This instrument produces signals in various forms, with adjustable amplitude and frequency.
- Oscilloscope: The oscilloscope measures the voltage and frequency of electrical signals, offering a visual representation of these signals on its display.
- Multimeter: This is a fundamental measuring tool used to assess various electrical properties.
- Breadboard: Also known as a protoboard, this serves as a construction base for assembling electronic circuits.

Each of these tools plays a vital role in the design, analysis, and troubleshooting of electronic circuits.



(a) Power Supply



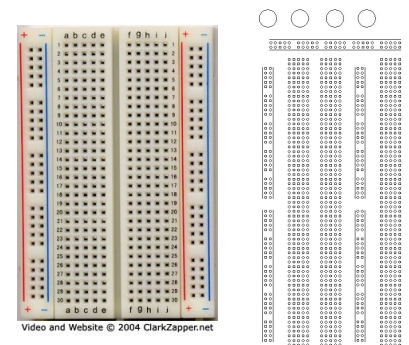
(b) Function Generator



(c) Oscilloscope



(d) Multimeter



(e) Breadboard

Fig. 1.1 Equipment in Electronic Laboratory

Each workstation in the lab has the necessary equipment: an oscilloscope, a power supply, an analog multimeter and an experimental box. The multimeter can measure voltage, current, resistance and capacitance. The experimental box includes ± 5 V and ± 12 V power supplies, breadboard, and a function generator, which is sufficient for experiments.

The breadboarding area, as shown in Fig. 1.2, on the experimental box has holes for component leads and IC pins. The right size of wires are needed otherwise the board will be ruined. The horizontal rows of holes on the top and bottom of the breadboard are connected together horizontally. The left and right halves are independent. We suggest that you use these horizontal rows for power supply voltages and ground. You may wish to put a jumper wire between the left and right halves so that the voltages are the same across the board. The vertical columns of holes are connected electrically in groups of five along a vertical line. The top and bottom halves are independent. Typically one inserts an IC chip straddling the center through.

There are then four empty holes for making connections to each IC pin. When you plug IC's into the breadboard, a common convention is to put pin 1 on the left. For other components, make sure the leads are not in the same column of five unless you want them connected together.

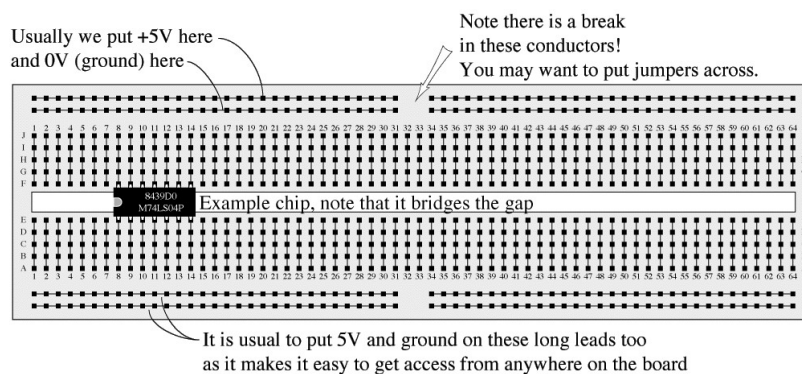


Fig. 1.2 The Breadboard Area

The oscilloscope has many knobs and buttons which may be confusing at first. It helps if you read the introductory booklet and manual provided by Tektronix. If nothing seems to be happening press AUTOSET.

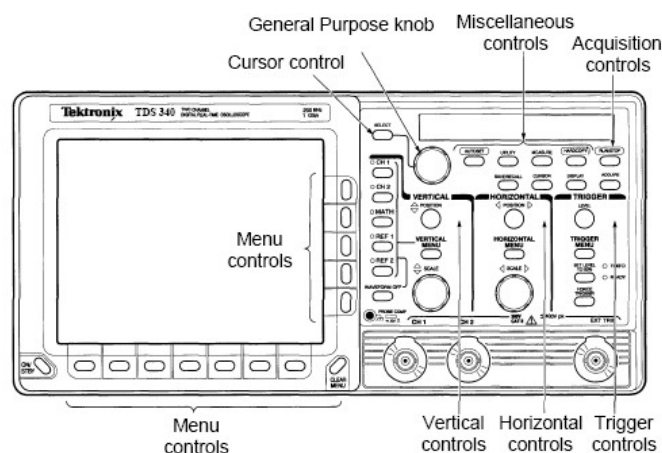


Fig. 1.3 Oscilloscope

Lab 1: Introduction to Electronics Laboratory

Apparatus List		Quantity
1.	Diode: 1N4001	1
2.	Resistors:	
	1 k Ω , 4.7 k Ω , 10 k Ω , 22 k Ω	1 each
3.	Power supply	1
4.	Multimeter	1

Characteristics of PN junction diode

Lab procedure

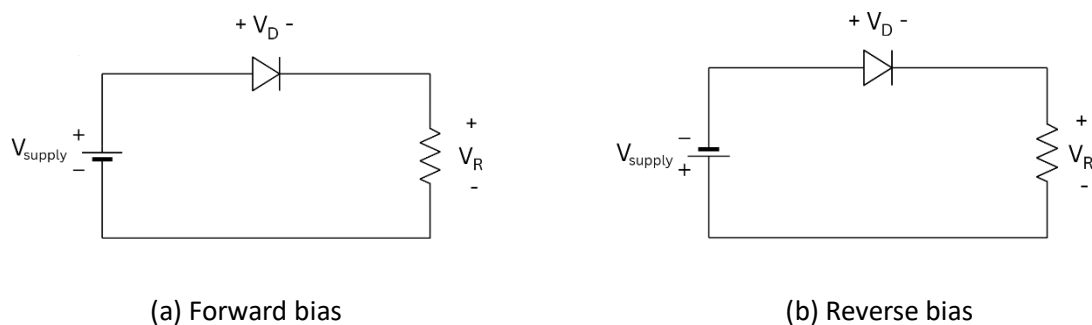


Fig. 1.4 Diode Circuits

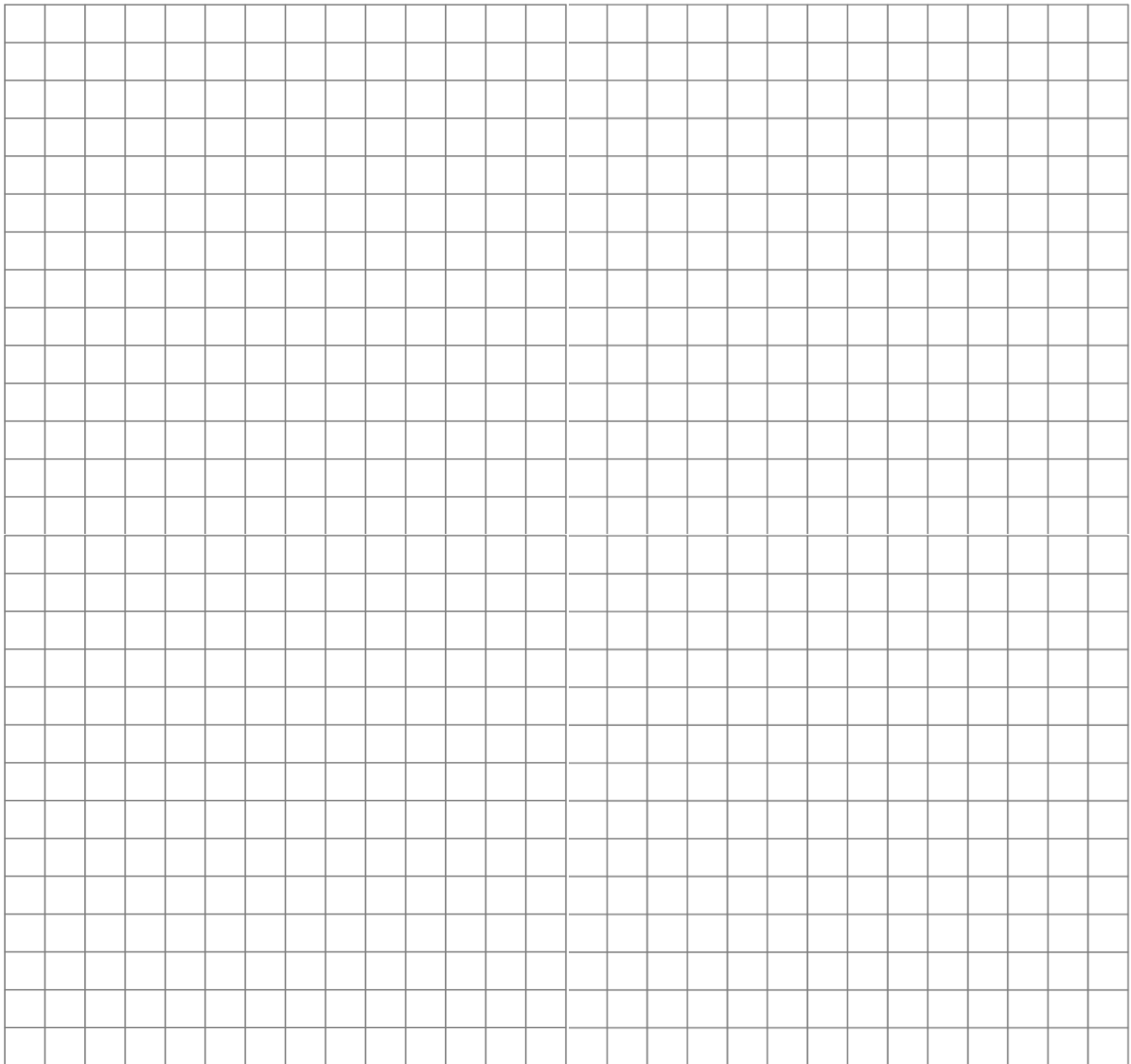
1. Build a forward biased diode circuit as shown in Fig.1.4(a).
2. Measure voltage across the diode and voltage across the resistor.
3. Repeat the previous procedure with different resistors according to Table 1.1.
4. Build a reverse biased diode circuit as shown in Fig.1.4(b).
5. Measure voltage across the diode and voltage across resistor.
6. Repeat the previous procedure with different resistors.
7. Plot I-V characteristic curve of diode from the experiment results.

Lab result

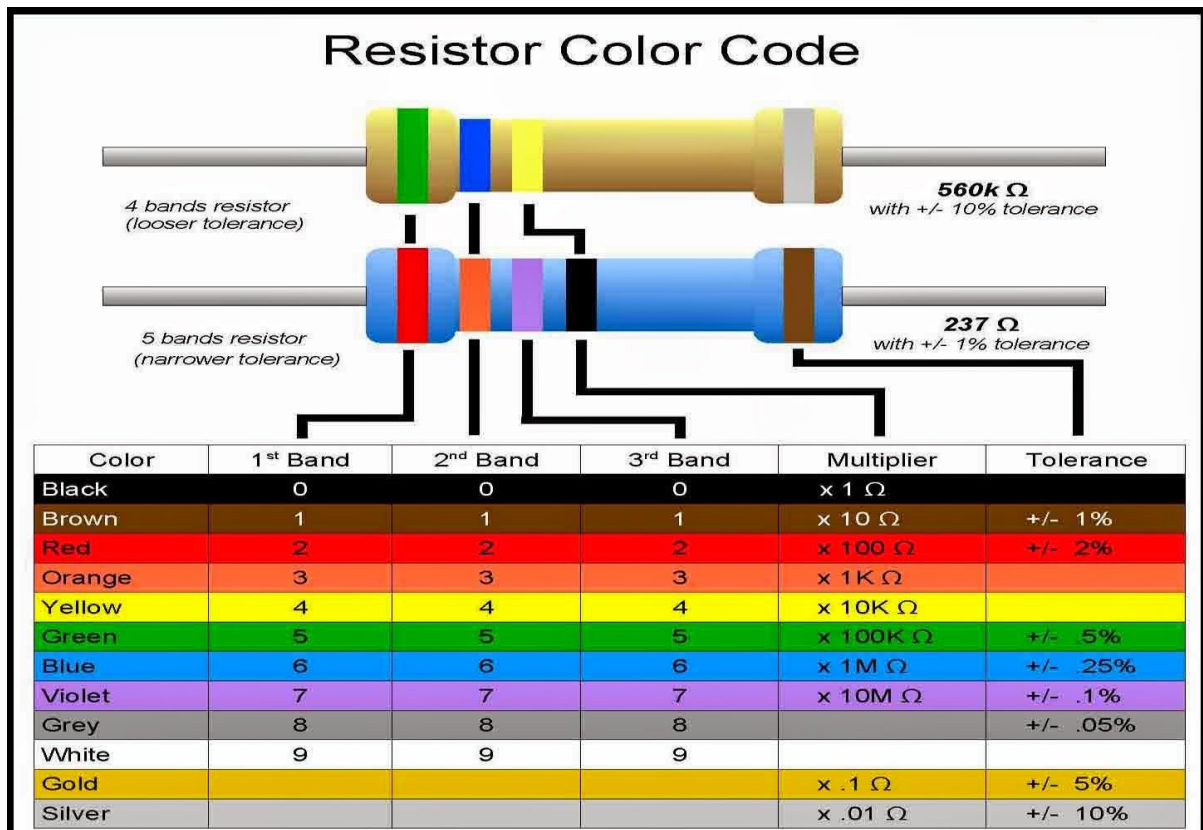
Table 1.1

	R(Ω)	22k	10k	4.7k	1k
Forward Bias	V_D				
	V_R				
	I_D				
Reverse Bias	V_D				
	V_R				
	I_D				

I-V Characteristic Curve



References



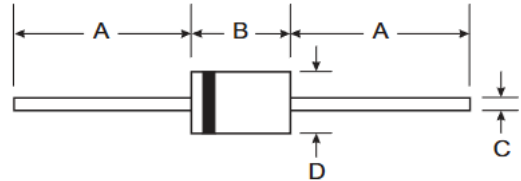
[Resistor Color Code Calculator | Tools - Utmel](#)

Features

- Diffused Junction
- High Current Capability and Low Forward Voltage Drop
- Surge Overload Rating to 30A Peak
- Low Reverse Leakage Current
- Lead Free Finish, RoHS Compliant (Note 4)

Mechanical Data

- Case: DO-41, A-405
- Case Material: Molded Plastic. UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020C
- Terminals: Finish - Bright Tin. Plated Leads Solderable per MIL-STD-202, Method 208
- Polarity: Cathode Band
- Mounting Position: Any
- Ordering Information: See Last Page
- Marking: Type Number
- Weight: DO-41 0.30 grams (approximate)
A-405 0.20 grams (approximate)



Dim	DO-41 Plastic		A-405	
	Min	Max	Min	Max
A	25.40	—	25.40	—
B	4.06	5.21	4.10	5.20
C	0.71	0.864	0.53	0.64
D	2.00	2.72	2.00	2.70

All Dimensions in mm

"L" Suffix Designates A-405 Package
No Suffix Designates DO-41 Package

Maximum Ratings and Electrical Characteristics @ $T_A = 25^\circ\text{C}$ unless otherwise specified

Single phase, half wave, 60Hz, resistive or inductive load.
For capacitive load, derate current by 20%.

Characteristic	Symbol	1N 4001/L	1N 4002/L	1N 4003/L	1N 4004/L	1N 4005/L	1N 4006/L	1N 4007/L	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	50	100	200	400	600	800	1000	V
RMS Reverse Voltage	$V_R(RMS)$	35	70	140	280	420	560	700	V
Average Rectified Output Current (Note 1) @ $T_A = 75^{\circ}C$	I_O	1.0							A
Non-Repetitive Peak Forward Surge Current 8.3ms single half sine-wave superimposed on rated load (JEDEC Method)	I_{FSM}	30							A
Forward Voltage @ $I_F = 1.0A$	V_{FM}	1.0							V
Peak Reverse Current @ $T_A = 25^{\circ}C$ at Rated DC Blocking Voltage @ $T_A = 100^{\circ}C$	I_{RM}	5.0 50							μA
Typical Junction Capacitance (Note 2)	C_j	15				8			pF
Typical Thermal Resistance Junction to Ambient	$R_{\theta JA}$	100							K/W
Maximum DC Blocking Voltage Temperature	T_A	+150							$^{\circ}C$
Operating and Storage Temperature Range (Note 3)	T_j, T_{STG}	-65 to +150							$^{\circ}C$

- Notes:
1. Leads maintained at ambient temperature at a distance of 9.5mm from the case.
 2. Measured at 1. MHz and applied reverse voltage of 4.0V DC.
 3. JEDEC Value.
 4. RoHS revision 13.2.2003. Glass and High Temperature Solder Exemptions Applied, see *EU Directive Annex Notes 5 and 7*.

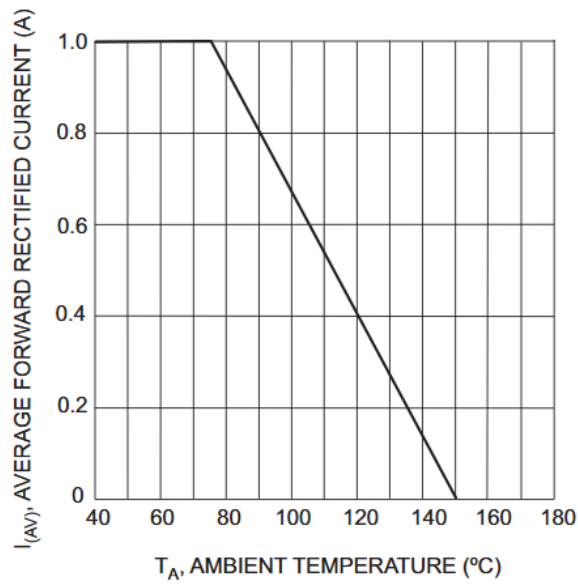


Fig. 1 Forward Current Derating Curve

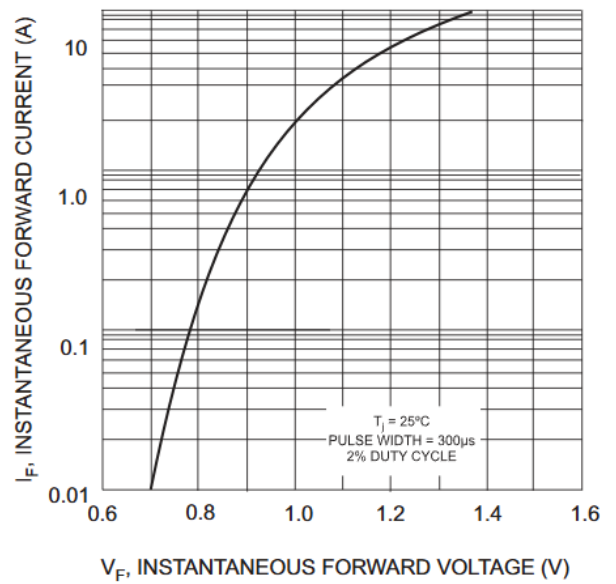


Fig. 2 Typical Forward Characteristics

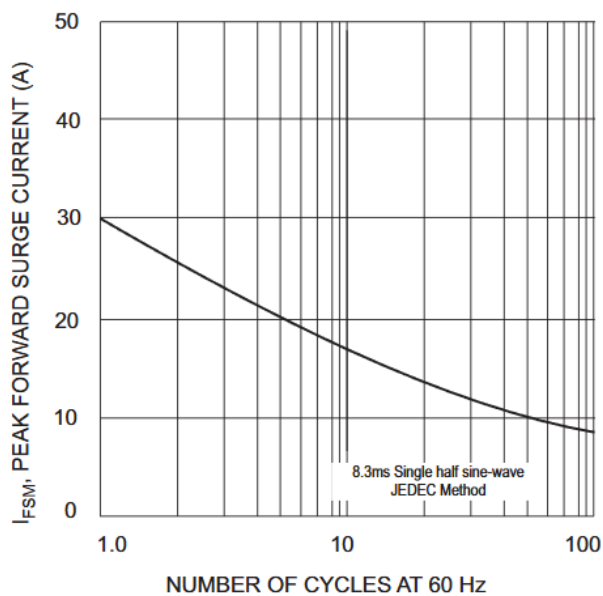


Fig. 3 Max Non-Repetitive Peak Fwd Surge Current

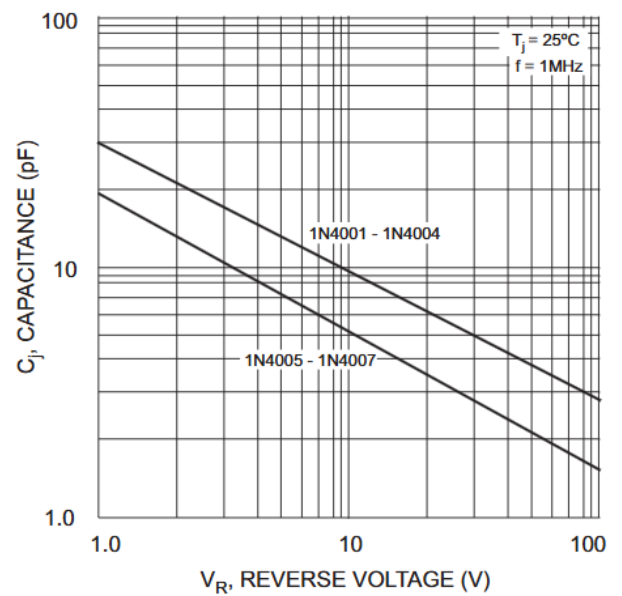


Fig. 4 Typical Junction Capacitance